



Yiwei Shu

📍 Beijing, China ✉️ shuyiwei24@mails.ucas.ac.cn 🔗 shu00111.github.io 🌐 Shu00111

Education

Ph.D.	University of Chinese Academy of Sciences , Robotics and Space Robotics - Affiliated with the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences. - Research on autonomous exploration, legged field robots, active SLAM, motion planning, and planetary robotics.	Beijing, China 2024 – present
B.Eng.	ShanghaiTech University , Electronic Information Engineering Undergraduate research in robotics, visual control, point-cloud registration, medical imaging, and integrated-circuit design.	Shanghai, China Sept 2020 – June 2024
Exchange	University of Illinois Urbana-Champaign , Exchange Program Completed Quantum Physics, Data Science Discovery, and Statistical Analysis with A+ grades.	Urbana, IL, USA Dec 2022 – May 2023

Manuscripts Under Review

Finite-State Visual Servo Control for Autonomous Target Landing of Low-Cost Underactuated Fixed-Wing Aircraft Yiwei Shu, Shuquan Wang. Manuscript under review at <i>Aerospace Science and Technology</i> .	2026
From End-Effector Targets to Whole-Arm Postures: Configuration-Space Waypoint Recovery for Reactive Manipulation Yiwei Shu et al. Manuscript submitted to <i>IEEE Robotics and Automation Letters</i> (submission status: received).	2026
RKILPlanner: Risk-based Kinematic Imitation Learning Planning Towards Human-Like Driving in Highway Scenarios Yiwei Shu, Shuquan Wang. Manuscript submitted to <i>ROBIO 2026</i> .	2026

Publications

Expanding Grasp Reachability: Field-based Pose-constrained Two-stage Grasp Planning Yiwei Shu, Shuquan Wang, Jun Li, Hanying Sang, and Yidong Ye. Accepted full paper and oral presentation at the <i>2026 5th International Conference on Service Robotics (ICoSR 2026)</i> .	2026
Learning-based Extended-workspace Manipulation: A Self-relocating Space Crawler for Autonomous Deep-space Outposts Yiwei Shu, Shuquan Wang. Poster presentation at the <i>46th COSPAR Scientific Assembly</i> .	2026
Attitude Control of Unpowered Aircraft in Flight Based on Kalman Filter Visual Servo-PID Optimization Algorithm Yiwei Shu, Shuquan Wang. Accepted for presentation and publication in the <i>Proceedings of the 44th Chinese Control Conference (CCC 2025)</i> .	2025

Projects

Time-Efficient Multi-Gait Lunar Navigation for Unitree Go2 Developing a goal-directed SLAM and adaptive locomotion framework for fast, reliable quadruped navigation across GPS-denied lunar terrain. - Integrate Livox Mid-360 LiDAR-inertial odometry, occupancy-grid mapping, and point-goal planning to navigate the Unitree Go2 toward specified lunar targets. - Design a hybrid locomotion planner that switches between walking and jumping based on terrain traversability, shortcut gain, and execution risk, minimizing time-to-goal across craters, gaps, and ridges. - Build 0.16g Isaac Lab lunar-terrain benchmarks and gait-conditioned world models to predict traversal cost, observation gain, and localization drift, evaluating time-to-goal, path length, success rate, and sim-to-real robustness.	2026 – present
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Technical Skills

Programming: Python, MATLAB, C, C++, R, Verilog, LaTeX

Robotics: ROS/ROS 2, PCL, Unitree Go2, RM75, STM32, LiDAR and depth sensing

Simulation and Learning: Isaac Lab, MuJoCo, PyTorch, scikit-learn, numerical simulation

Engineering Tools: Git, Linux, SolidWorks, Altium Designer, ANSYS Fluent, Cadence, Vivado

Selected Honors

- Outstanding Student Award, University of Chinese Academy of Sciences (2025-2026)
- Excellence Rising Star Award, UCAS Student Union (2025-2026; 2024-2025)
- Outstanding Student Leader Award, University of Chinese Academy of Sciences (2024-2025)
- Third Prize, RoboMaster National Competition (2022)
- Second Prize, East China Division, 6th National Undergraduate IC Innovation and Entrepreneurship Competition (2022)

Teaching and Leadership

- Department Head, Student Life and Culture, UCAS Student Union (2024-Present): lead department operations and coordinate campus cultural and student-life initiatives.
- Teaching Assistant, Digital Circuits, ShanghaiTech University (September-December 2022): led tutorials and assisted with grading.
- Sentry Team Lead, RoboMaster (2021-2023): coordinated embedded programming, PCB design, mechanical modeling, and team planning.
- Volunteer Teacher, ShanghaiTech University Social Practice Program, Yunlong, Yunnan (Jul 2021): taught at a local primary school and conducted field research on local education conditions.